Effectime Enterprise

Software Valuation & Technical Due-Diligence Report

Software Valuation & Technical Due-Diligence Report

Date / Dátum:	2026-05-11
Repository:	HenrikFaul/effectime-app-enterprise-a95029a1
Version:	v2.6.0+
Language:	English
Confidence:	Medium-High / Közepes-Magas

Table of Contents

1. Executive Summary

Effectime Enterprise is a **multi-tenant, enterprise-grade HR leave management and workforce planning SaaS platform** built for the Hungarian/CEE SME and mid-market segment. It provides end-to-end leave lifecycle management, multi-view calendar with real-time timeline, resource and project capacity planning, Agile integration (Jira + Azure DevOps), custom report builder with SQL mode, and mobile support via Capacitor.

Key Statistics

170	31,435	53	17	77	2 months
Source Files (.ts/.tsx)	Lines of Code	DB Migrations	Edge Functions	Enterprise Components	Dev Timeline

Build Effort Summary

Metric	Low	Most Likely	High
Person-hours	2,041	3,051	4,485
Person-months (160 h/mo)	12.8	19.1	28.0
Calendar months (3-person team)	5.5 mo	7.5 mo	11 mo

Build Cost Summary

Scenario	Low	Most Likely	High
CEE / Hungary team	€118,000	€178,000	€245,000
Western EU agency	€265,000	€380,000	€520,000
Mixed CEE + WEU lead	€165,000	€235,000	€330,000

Market Value Summary

Scenario	Estimated Range	Central Estimate
Pre-revenue IP sale	€200K – €600K	€380K
Early customers (€50–100K ARR)	€400K – €1,200K	€700K
Growth stage (€300–500K ARR)	€1.8M – €4.5M	€3.0M
Strategic acquisition	€700K – €2,500K	€1.5M

Scenario	Estimated Range	Central Estimate
Current state (unknown revenue)	€350K – €900K	€550K

■ **Biggest Uncertainty Driver:** Revenue/ARR data is not available in the repository. This single factor is the most dominant variable in market valuation. A €100K ARR trajectory would move the central estimate from €550K to ~€700–1,000K.

2. Product Reconstruction

Technical Architecture

Layer	Technology	Version
Frontend Framework	React + TypeScript + Vite	18.3 / 5.8 / 5.4
UI Components	shadcn/ui (Radix UI) + Tailwind CSS	3.4.17
Animation	Framer Motion	12.35
State / Data Fetching	TanStack Query	v5.83
Charts	Recharts	2.15
Virtualization	@tanstack/react-virtual	3.13
Drag & Drop	@dnd-kit	6.3
Mobile	Capacitor (iOS + Android)	8.2
Forms / Validation	React Hook Form + Zod	7.6 / 3.25
Backend / DB	Supabase (PostgreSQL)	2.98
Edge Functions	Deno (TypeScript)	17 functions
Auth	Supabase Auth + Google OAuth	Custom activation
Agile Integration	Jira REST API v3 + Azure DevOps WIQL	Bidirectional
Email	Lovable Email API (Mailgun)	Transactional
Scheduling	pg_cron	2 jobs

Feature Module Map

Module	Key Capabilities
Authentication & Identity	Google OAuth + custom activation gate, OTP email, account deletion (GDPR), admin panel
Workspace & Tenancy	Multi-workspace, invitation system, instant member creation, role management, branding
Leave Request Lifecycle	6-status state machine, 2-step submit, substitute picker, attachment upload, quota tracking, admin override
Conflict Engine	Parallel validation: holidays, blocked dates, daily rules, office coverage, self-overlap (lib/conflictEngine.ts)
Capacity Engine	Hours-based: base_working_hours × allocation%, leave deduction, project allocation (lib/capacityEngine.ts)
Calendar & Timeline	3 views: monthly / Absentify-timeline (TanStack Virtual) / annual grid; dynamic filter bar with drag&drop;

Module	Key Capabilities
Coverage Planner	Weekly/monthly requirement vs. supply; smart batch scheduling algorithm; skill/role matching
Approval Chain & Escalation	Multi-step configurable chains, escalation rules with hour threshold, bulk approve/reject
Resource Management	Projects + Gantt + capacity gap + utilization heatmap + scenario planner + financials
Agile Integration	Jira + ADO: backlog browser, issue writeback, capacity fit, what-if simulation, field discovery
Report Builder	Visual config + SQL mode + live preview + scheduled delivery; pinned report widgets
Email Infrastructure	Auth hook, transactional templates, email queue, suppression/unsubscribe, scheduled reports
Notifications	In-app notifications, read/unread, preferences per event type, iCal subscription
Audit & Compliance	Immutable audit log, GDPR account deletion, CSV export, export job tracking

3. Scope Decomposition

Complexity Ratings by Area

Feature Area	Complexity	Hidden Cost Drivers
Auth & Identity	Medium	Custom Google OAuth activation gate (300+ lines backend logic); legacy bypass rules
Multi-Tenant Architecture	High	RLS policies on 30+ tables; schema split (syncfolk/plannermaster); dynamic permission tree
Leave Request Lifecycle	High	6-state machine; quota transaction scoping bug (fixed v2.6); substitute ordering
Conflict Engine	Medium-High	6-table parallel fetch; multi-position OR semantics; scalar/array column fallback
Capacity Engine	Medium	Hours math; allocation %; leave deduction; silent failure risk (fixed)
Calendar & Timeline	High	TanStack Virtual (200+ rows); coverage planner race conditions; filter persistence
Approval Chain	Medium	Multi-step config; escalation timers; role routing
Resource Management	High	Gantt drag-resize; scenario planner; financials; utilization heatmap
Agile Integration	Very High	Jira REST v3 + ADO WIQL in one proxy; empty patch protection; base-URL normalisation
Report Builder	High	SQL mode with workspace-scoped access control; live preview; scheduled cron delivery
Email Infrastructure	High	Auth hook intercept; suppression/bounce; React Email templates; queue retry logic

Complexity Multipliers

Factor	Impact	Rationale
Hungarian-only UI	+0%	No i18n framework; hardcoded strings (tech debt, not build cost)
Multi-schema PostgreSQL	+15%	3 schemas; helper functions; non-destructive migration constraint
Dual-product codebase	+10%	Syncfolk + Enterprise share infrastructure; coordination overhead
AI-accelerated development	–30%	Build time reflects AI tooling; traditional estimate 2–3× longer
Minimal automated tests	+20%	Manual QA burden; regression risk higher; bug-fix cycles needed

Factor	Impact	Rationale
17 Deno edge functions	+15%	Cold start considerations; Deno vs Node API differences; cold-start latency

4. Methodology

Bottom-Up Estimation

Each of 10 major areas decomposed into sub-components. Effort estimated per component based on: lines of code, DB tables, API endpoints, UI interaction complexity, edge cases evidenced in changelog.

Analogous Estimation

Comparable products (Calamari, Timetastic, Sloneek, Absence.io) used as reference points. Public engineering blog posts from HR SaaS startups confirm 6–18 month MVP timelines.

Three-Point PERT

Optimistic / Most Likely / Pessimistic estimates per area. Formula: $E = (O + 4M + P) / 6$. Standard deviation: $SD = (P – O) / 6$.

Code Volume Baseline

31,435 lines ÷ 120 LOC/day average = 262 developer-days. × 2.2 overhead multiplier (QA, PM, design, review, meetings, rework) = ~576 days = ~2,880 person-hours. Aligns with bottom-up estimate.

Important Distinctions

- **Effort ≠ Duration:** 2,880 person-hours can be delivered in 5 months (3 FTE) or 24 months (0.5 FTE).
- **Build Cost ≠ Market Value:** Replacement cost is the floor; revenue traction and strategic positioning multiply it significantly.
- **Code ≠ Product:** A shipped product includes design decisions, user research, content strategy, and customer success not visible in code.
- **AI-assisted ≠ Traditional:** The 2-month build reflects Lovable/Claude tooling. Conventional teams take 2–3× longer.

5. Team Composition

Required Roles

Role	Why Needed	Phase	Effort %
Senior Full-Stack Engineer	Architecture, Supabase schema, RLS, edge functions, business logic	All	35%
Mid-Level Frontend Engineer	React components, Recharts, DnD Kit, Framer Motion, responsive UI	2–8	30%
Backend / DB Engineer	PostgreSQL schema, RLS policies, SECURITY DEFINER functions, migrations	1–3, 6–8	15%
UX/UI Designer	Information architecture, component library, mobile layouts, data viz	1–2, 4–5	10%
Product Manager	Requirements, sprint planning, backlog prioritisation, stakeholder mgmt	All	7%
QA Engineer	Manual testing, regression, cross-browser/device	3–8	3%

Delivery Team Options

Lean Team (3 people, 7–10 months)
1× Senior Full-Stack (lead, 100%) · 1× Mid Frontend (100%) · 0.5× UX/UI Designer (50%)
Lowest cost; PM/QA done by lead + stakeholder. Best for a solo founder or bootstrapped product.

Balanced Team (5 people, 5–7 months)
1× Senior Full-Stack · 1× Mid Frontend · 1× Backend/DB · 0.5× UX/UI · 0.5× PM
Recommended for a seed-stage startup. Parallel tracks on frontend/backend. Reasonable velocity.

Enterprise Delivery Team (8 people, 4–5 months)
2× Senior · 2× Mid Frontend · 1× Backend/DB · 1× UX/UI · 1× PM · 1× QA
Maximum velocity. Suitable for agency delivery or well-funded sprint. Highest coordination overhead.

6. Effort Estimate

Area-by-Area Breakdown (PERT)

Feature Area	Opt. (h)	ML (h)	Pess. (h)	PERT (h)
Auth & Identity	80	120	180	122
Workspace & Tenancy	120	180	260	183
Leave Request Lifecycle	180	260	380	263
Conflict Engine	60	90	140	92
Capacity Engine	50	75	120	77
Calendar & Timeline Views	200	290	420	293
Approval Chain & Escalation	60	90	140	92
Resource Management	180	260	380	263
Agile Integration	140	200	300	203
Report Builder & Reporting	120	180	270	183
Email Infrastructure	90	130	200	133
Admin, Profile, Landing	60	90	140	92
Mobile (Capacitor) + Polish	60	90	130	90
DB Schema + RLS + Migrations	80	120	180	122
Smart Schedule Algorithm	40	60	90	61
Permission System (dynamic tree)	50	75	120	77
Subtotal (coding)	1,570	2,310	3,450	2,347
+ 30% overhead (QA, PM, design, deploy)	471	693	1,035	704
TOTAL	2,041	3,003	4,485	3,051

Summary in Multiple Units

Metric	Low	Most Likely	High
Person-hours	2,041	3,051	4,485
Person-days (8h)	255	381	561
Person-months (160h)	12.8	19.1	28.0
Calendar months (3-person core)	5.5	7.5	11
Calendar months (5-person team)	3.5	5.0	7.0

7. Cost Estimate

CEE / Hungary Rate Assumptions (2025–2026)

Role	Junior (€/h)	Mid (€/h)	Senior (€/h)
Full-Stack Engineer	€15–22	€28–40	€42–65
Frontend Engineer	€14–20	€25–38	€38–60
Backend / DB Engineer	€15–22	€28–42	€42–65
UX/UI Designer	€14–20	€24–36	€36–55
Product Manager	€18–25	€32–45	€45–70
QA Engineer	€12–18	€22–32	€32–48

Detailed Cost Model — Most Likely (CEE Rates)

Role	Share	Hours	Rate	Cost
Senior Full-Stack Engineer	35%	1,068	€52/h	€55,536
Mid-Level Frontend Engineer	30%	915	€33/h	€30,195
Backend / DB Engineer	15%	458	€48/h	€21,984
UX/UI Designer	10%	305	€30/h	€9,150
Product Manager	7%	214	€38/h	€8,132
QA Engineer	3%	91	€28/h	€2,548
Direct Labour		3,051		€127,545
Overhead (25%)				€31,886
Contingency (15%)				€19,132
TOTAL				€178,563

Cost Ranges by Scenario

Scenario	Low	Most Likely	High
CEE / Hungary (lean team)	€118,000	€178,000	€245,000
Western EU agency	€265,000	€380,000	€520,000
Mixed CEE + WEU lead	€165,000	€235,000	€330,000

8. Market Comparison

Comparable Products

Product	HQ	Price/User/Mo	Key Notes vs. Effectime
Calamari	Poland	€2.50–5.00	Closest CEE peer; no Agile integration
Timetastic	UK	$1.50–2.50	Much simpler; acquired 2023; $1.7M ARR
Absence.io	Germany	€3.50–5.00	EU-focused, GDPR; similar scope
Factorial HR	Spain	€5–8	Broader HR (payroll); less Agile depth
Personio	Germany	€5–25	Enterprise, payroll; €8.5B valuation peak
Sloneek	Czech/CEE	~€5–10 est.	Closest CEE; Hungarian UI; €3.6M raised 2024
BambooHR	USA	$10–25 est.	North American SME; no Agile; strong brand
Float	Australia	$6–12	Resource planning only; no leave management
Teamdeck	Poland	$3.99	Resource + leave; closest overall comparable

HR SaaS Valuation Multiples (2024–2026)

Growth Rate	Typical ARR Multiple	Examples
>100% YoY (early stage)	10–15× ARR	Rippling: 29×; HiBob: 22×
50–100% YoY	7–10× ARR	Personio peak: 24×
20–50% YoY (mature SaaS)	5–8× ARR	Deputy: ~11× (2022)
<20% YoY (stable)	3–5× ARR	Bootstrapped SaaS median: 4.8×
M&A; median (2024)	4.1× ARR	SaaS Capital Index
M&A; median (2025)	3.1–3.8× ARR	Current deal environment

Market Sizing

Segment	2024 Market Size	CAGR	Notes
Global HR Software	$20.5 billion	10.1%	To 2032
Absence & Leave Management	~$1.2 billion	9.5%	To 2035
European Workforce Management	$2.66 billion	8.67%	To 2030
CEE HR Tech	~€200–250M	~12–15%	Estimate

9. Market Value Estimate

Five valuation lenses are applied and triangulated to produce a range-based estimate.

Lens 1: Replacement / IP Value

Build cost at CEE rates: €145K–€245K. As marketed IP asset, 1.5–2.5× multiplier applied. Result: **€200,000 – €550,000** (floor value — zero revenue assumed).

Lens 2: Comparable-Based Valuation

At 5–8× ARR multiples (current market): €30K ARR → €240–400K; €100K ARR → €700K–1M; €500K ARR → €2.5–4M. Sloneek (closest CEE peer) raised €3.6M at ~€5–9M ARR.

Lens 3: Feature Depth Premium

Agile integration (Jira+ADO bidirectional), SQL report builder, smart scheduling, multi-schema architecture — features exceeding typical price-tier competitors. Premium: 1.3–1.8× vs. IP value → **€260,000 – €990,000**.

Lens 4: Risk Adjustment

Risk haircuts: minimal test coverage (–15%), AI-assisted quality variability (–10%), Hungarian-only UI (–10%). Offsetting: modern stack (+10%), clean RLS (+5%), active maintenance (+8%). Net: –12% adjustment.

Lens 5: Strategic Acquisition Premium

Strategic acquirer (CEE HR roll-up, Jira-adjacent vendor) would pay 1.5–3× IP value for the agile integration module, enterprise workflow engine, or CEE market entry. Range: **€300,000 – €1,650,000**.

Final Market Value Ranges

Scenario	Range	Central Estimate
Pre-revenue IP sale	€200K – €600K	€380K
Early customers (€30–100K ARR)	€400K – €1,200K	€700K
Post-PMF (€200–500K ARR)	€1.5M – €4.5M	€3.0M
Strategic acquisition	€700K – €2,500K	€1,500K
Current state (unknown revenue)	€350K – €900K	€550K

Central Estimate: €550,000 — assuming early traction, pre-growth stage, unknown revenue. A demonstrated €100K ARR trajectory would revise this to ~€700–1,000K. Strategic acquisition interest from a CEE HR tech player could push to €1–2M.

10. Assumptions and Limitations

Hard Evidence (Known)

✓ Full source code: 170 files, 31,435 lines of TypeScript/TSX
✓ 53 database migration files with complete schema history
✓ Complete changelog from v2.0.0 to v2.6.0+ with Jira ticket references
✓ 17 Supabase edge functions with full implementation
✓ All npm dependencies with versions (60+ packages)
✓ Development timeline: 2026-03-07 to 2026-05-11 (~2 months)
✓ Governance documentation and versioning artifacts
✓ Architecture patterns: Supabase + React + TypeScript + Deno

Inferred (Not Directly Observed)

~ Development effort (estimated from code volume, complexity signals, comparables)
~ Rate assumptions (CEE market research, industry norms)
~ Market comparables (public pricing pages, press releases)
~ Revenue potential (pricing model based on comparable SaaS)
~ Team composition (inferred from codebase breadth and complexity)

Unknown / Missing

✗ Revenue / ARR — no monetisation data; most dominant valuation variable
✗ User/tenant count — no telemetry or analytics integration visible
✗ Customer data — no CRM, support tickets, or user feedback data
✗ Payroll integration — absent; limits enterprise HR-suite evaluations
✗ SLA / uptime history — no monitoring or incident data
✗ Mobile app store status — Capacitor configured but no evidence of published apps

11. Recommended Next Steps

For Cost Optimisation

→ Add automated test coverage (target 70%+ on conflictEngine, capacityEngine, smartSchedule)
→ Replace 4-second DB polling with Supabase Realtime subscriptions
→ Implement audit log as queue-backed system (not fire-and-forget)
→ Introduce i18next for proper internationalisation (enables non-Hungarian markets)
→ Resolve (supabase as any) type casts in conflict engine

For Product Sale / Acquisition / Fundraising

→ Establish measurable ARR (even €5–20K early revenue dramatically increases valuation credibility)
→ Publish English-language version of product and landing page
→ Obtain at least one referenceable enterprise customer (50+ employees)
→ Commission security audit of RLS policies and edge function authentication
→ Prepare data room: financials, technical architecture doc, customer list, IP ownership chain

For Roadmap Prioritisation

→ Payroll integration (Nexon/HR365 for CEE) — largest enterprise sales barrier
→ Microsoft Teams notification connector — expands addressable market significantly
→ SCIM/SSO — required for large enterprise procurement
→ Protect Jira/ADO agile integration moat — no direct competitor at this price tier

For Technical Debt Reduction

→ Priority 1: Automated test suite (vitest configured but unused)
→ Priority 2: Audit log reliability (fire-and-forget is a compliance risk)
→ Priority 3: Replace polling with Realtime subscriptions
→ Priority 4: i18n refactor for internationalisation
→ Priority 5: Type-safe Supabase queries (remove (as any) casts)

12. Appendix

A. Database Table Inventory (Selected)

Table	Module	Purpose
enterprise_workspaces	Workspace	Tenant workspace records
enterprise_memberships	Workspace	User-workspace relations + base_working_hours
enterprise_invitations	Workspace	Email-based invitations with token expiry
leave_requests	Leave	6-status leave request records
approval_decisions	Leave	Approval/rejection decisions per step
leave_request_substitutes	Leave	Ordered substitute assignments
enterprise_leave_types	Config	Custom leave type definitions
enterprise_holidays	Config	Workspace holiday calendar
enterprise_office_coverage_rules	Config	Multi-position/skill staffing rules
enterprise_approval_chains	Config	Multi-step approval definitions
enterprise_audit_events	Audit	Immutable audit trail (prev/new state JSON)
enterprise_agile_issues	Agile	Cached Jira/ADO issue records
enterprise_feature_catalog	Permissions	Recursive permission tree (parent_key)
enterprise_quota_transactions	Quota	Leave quota debit/credit ledger
tenant_calendar_settings	Calendar	Per-workspace filter config (JSONB)

B. Confidence Assessment

Dimension	Confidence	Rationale
Build effort estimate	High (±20%)	Full codebase available; changelog evidence
Cost estimate (CEE)	High (±25%)	Market rate data available
Cost estimate (WEU)	Medium (±35%)	Indirect evidence only
Market value (pre-revenue)	Medium (±40%)	No revenue; comparable-based
Market value (with revenue)	Medium-High (±25%)	Standard ARR multiples applicable

C. Comparable Acquisitions

Target	Acquirer	Year	Price	Notes
ReedGroup	Alight Inc.	2022	$87M	Absence mgmt, 50% Fortune 100
AbsenceSoft	Luminate Capital	2024	Undisclosed	VC-backed leave mgmt
Timetastic	Citation Group	2023	Undisclosed	$1.7M ARR; bootstrapped
WorkForce Software	ADP	2024	Undisclosed	Major workforce platform

This report was produced by AI-assisted technical due-diligence combining direct repository inspection and external market research. It is for informational purposes and does not constitute financial advice. Report Version 1.0 — 2026-05-11.